$\frac{...10^{-6} \times 2 \times 10^{-3}}{...0^{-4}} = 4 \times 100^{-11}$

$4\pi \times 10^{-4} \, \omega/m^2$

... produces an electric ... field.

... (ie) ...tion electric current produce ...tion of charges is at a ...ime, then steady current flows ...d by steady current i.e.

...(B)

...ey of force (mag flux) passing ...a plane at right angles to ...lled magnetic flux density

Relationship b/w $\bar{B}$ & $\bar{H}$

In magnetostatics $\bar{B}$ & $\bar{H}$ are related to each other through the property of region in which current carrying conductor is placed.

$$B = \mu H.$$

$$B = \mu_0 \mu_r H.$$

Biot savart's law

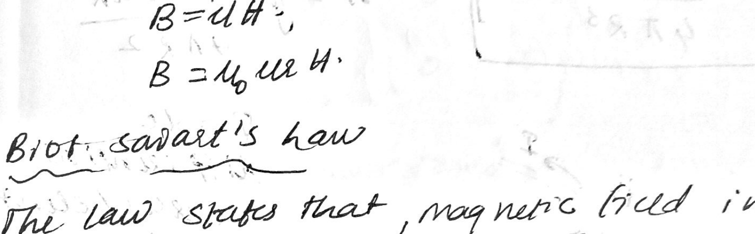
The law states that, magnetic field intensity $d\bar{H}$ produced at a pt P due to a differential current element $I dL$ is

1. Proportional to the product of current I & differential length dL.
2. The sine of angle b/w element & line joining pt P to the element
3. Inversely proportional to the square of distance R b/w pt P and the element.

$$d\bar{H} \propto \frac{I dL \sin\theta}{R^2}$$

$$d\bar{H} = \frac{K I dL \sin\theta}{R^2} \qquad K = \frac{1}{4\pi}$$

$$= \frac{I dL \sin\theta}{4\pi R^2}$$



dL = Magnitude of vector length $d\bar{L}$

$\hat{a}_R$ = unit vector in the dir^n from differential current element to point P.

$\sin\theta = dL \sin\theta$

A/m. ; $\hat{a}_R = \frac{\bar{R}}{|\bar{R}|}$

$\bar{H} = \oint \frac{I\, d\bar{L} \times \hat{a}_R}{4\pi R^2}$

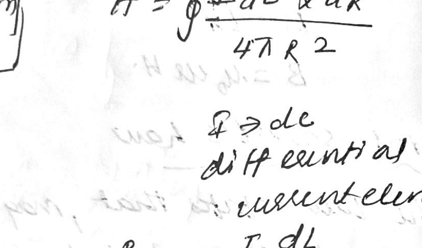
$I \Rightarrow dL$
differential current element
$I\, dL$

P

...avartz law.

long straight conductor:

$I\, d\bar{L} = I\, dz\, \hat{a}_z$



EC

$\bar{R}_{12} = -z\hat{a}_z + r\hat{a}_r$

$\hat{a}_{R12} = \frac{\bar{R}_{12}}{|\bar{R}_{12}|} = \frac{r\hat{a}_r - z\hat{a}_z}{\sqrt{r^2+z^2}}$

$$d\bar{L} \times \hat{a}_{R12} = \begin{vmatrix} \hat{a}_r & \hat{a}_\varphi & \hat{a}_z \\ 0 & 0 & dz \\ r & 0 & -z \end{vmatrix} = r\, dz\, \hat{a}_\varphi$$

$$I\, d\bar{L} \times \hat{a}_{R12} = \frac{I\, r\, dz\, \hat{a}_\varphi}{\sqrt{r^2+z^2}}$$

$$d\bar{H} = \frac{I\, d\bar{L} \times \bar{a}_{R12}}{4\pi R_{12}^2} = \frac{I\, r\, dz\, \hat{a}_\varphi}{4\pi\sqrt{r^2+z^2}\,(\sqrt{r^2+z^2})^2}$$

$$= \frac{I\, r\, dz\, \hat{a}_\varphi}{4\pi (r^2+z^2)^{3/2}}$$

$$\bar{H} = \int_{z=-\infty}^{\infty} d\bar{H} = \int_{-\infty}^{\infty} \frac{I\, r\, dz\, \hat{a}_\varphi}{4\pi (r^2+z^2)^{3/2}}$$

$z = r\tan\theta$, $z^2 = r^2\tan^2\theta$

$dz = r\sec^2\theta\, d\theta$, $z = -\infty$, $\theta = -\pi/2$ & $z = +\infty$, $\theta = +\frac{\pi}{2}$

$$H = \int_{\theta=-\pi/2}^{\pi/2} \frac{I\, r\, r\sec^2\theta\, d\theta\, \hat{a}_\varphi}{4\pi (r^2 + r^2\tan^2\theta)^{3/2}}$$

$$= \int_{\theta=-\pi/2}^{\pi/2} \frac{I\, r^2 \sec^2\theta\, d\theta\, \hat{a}_\varphi}{4\pi r^3 \sec^3\theta} \qquad 1+\tan^2\theta = \sec^2\theta$$

$$= \int_{\theta=-\pi/2}^{\pi/2} \frac{I}{4\pi r} \frac{1}{\sec\theta}\, d\theta\, \hat{a}_\varphi = \frac{I}{4\pi r} \int_{\theta=-\pi/2}^{\pi/2} \cos\theta\, d\theta\, \hat{a}_\varphi$$

$\pi/2 \quad \hat{a}_\varphi = \frac{I}{4\pi r}\left[\sin\pi/_2 - \sin(-\frac{\pi}{2})\right]\hat{a}_\varphi$

$\pi/2$

$-1)]\hat{a}_\varphi = \frac{2I}{2\times 4\pi r}\hat{a}_\varphi$

$\boxed{A/m}$

$\boxed{\hat{a}_\varphi \ wb/m^2}$

...nt conductor of finite lengths

The perpendicular distance of pt P from z axis is r. conductor placed at z axis. One end of conductor is at $z=z_1$ while other at $z=z_2$.

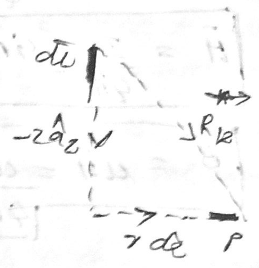
p point 2

...lement $d\bar{L}$ along z axis ...n Origin.

$= dz\,\hat{a}_z$

$$\hat{a}_{R12} = \frac{\bar{R}_{12}}{|\bar{R}_{12}|} = \frac{-z\hat{a}_z + r\hat{a}_r}{\sqrt{(-z)^2 + r^2}} = \frac{r\hat{a}_r - z\hat{a}_z}{\sqrt{r^2+z^2}}$$

$$d\bar{H} = \frac{I\,d\bar{L}\times\hat{a}_{R12}}{4\pi R_{12}^2}$$



$$d\bar{L}\times\hat{a}_{R12} = r\,dz\,\hat{a}_\varphi$$

$$I\,d\bar{L}\times\hat{a}_{R12} = \frac{I\,r\,dz\,\hat{a}_\varphi}{\sqrt{r^2+z^2}}$$

$$d\bar{H} = \frac{I\,r\,dz\,\hat{a}_\varphi}{4\pi\left(\sqrt{r^2+z^2}\right)\left(\sqrt{r^2+z^2}\right)^2}$$

$$= \frac{I\,r\,dz\,\hat{a}_\varphi}{4\pi(r^2+z^2)^{3/2}}$$

$$\bar{H} = \int_{z_1}^{z_2} d\bar{H} = \int_{z_1}^{z_2}\frac{Ir\,dz\,\hat{a}_\varphi}{4\pi(r^2+z^2)^{3/2}}$$

$z = r\tan\alpha, \quad z^2 = r^2\tan^2\alpha$

$dz = r\sec^2\alpha\, d\alpha$

for $z = z_1, \quad z_1 = r\tan\alpha_1$

for $z = z_2, \quad z_2 = r\tan\alpha_2$

$\alpha_1 = \tan^{-1}(z_1/r)$ & $\alpha_2 = \tan^{-1}(z_2/r)$

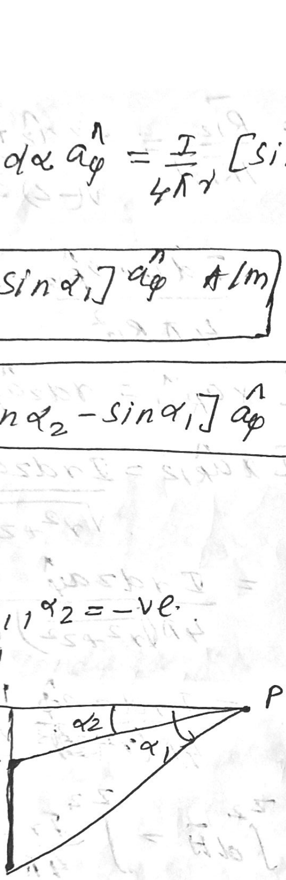
$$\bar{H} = \int_{\alpha_1}^{\alpha_2}\frac{I\,r\,r\sec^2\alpha\,d\alpha\,\hat{a}_\varphi}{4\pi\left[r^2 + r^2\tan^2\alpha\right]^{3/2}} = \int_{\alpha_1}^{\alpha_2}\frac{I\,d\alpha\,\hat{a}_\varphi}{4\pi\sec\alpha\, r}$$

$d\alpha\, \hat{a}_\varphi = \frac{I}{4\pi r}\left[\sin\alpha\right]_{\alpha_1}^{\alpha_2} \hat{a}_\varphi$

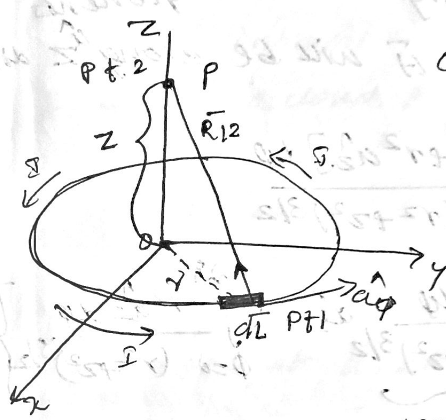
$\boxed{\ldots \sin\alpha_1]\, \hat{a}_\varphi \ \ A/m}$

$\boxed{\ldots n\alpha_2 - \sin\alpha_1]\, \hat{a}_\varphi \ \ wb/m^2}$

$\ldots, \alpha_2 = -ve.$



$\bar{H}$ on the axis of a circular loop



consider a circular loop carrying a direct current I, placed in xy plane with z axis as its axis

In cylindrical co-ordinate sim

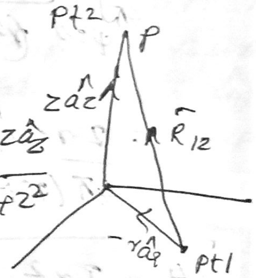
$$d\bar{L} = dr\hat{a}_r + r\,d\varphi\,\hat{a}_\varphi + dz\hat{a}_z$$

But $d\bar{L}$ is in the plane for which r is const. & $z=0$ const. plane.

$$I d\bar{L} = I\, r\, d\varphi\, \hat{a}_\varphi$$

$$\hat{a}_{R12} = \frac{\bar{R}_{12}}{|\bar{R}_{12}|} \qquad \bar{R}_{12} = -r\hat{a}_r + z\hat{a}_z$$

$$|\bar{R}_{12}| = \sqrt{(-r)^2 + z^2}$$



$$\hat{a}_{R12} = \frac{-r\hat{a}_r + z\hat{a}_z}{\sqrt{r^2+z^2}}$$

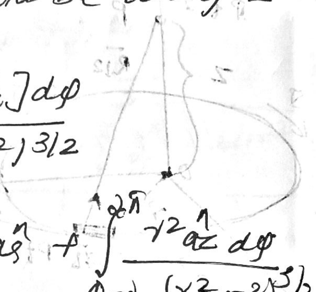
$$d\bar{L} \times \hat{a}_{R12} = \begin{vmatrix} \hat{a}_r & \hat{a}_\varphi & \hat{a}_z \\ 0 & r\,d\varphi & 0 \\ -r & 0 & z \end{vmatrix} = z\,r\,d\varphi\,\hat{a}_r + r^2\,d\varphi\,\hat{a}_z$$

$$d\bar{H} = \frac{I\, d\bar{L} \times \hat{a}_{R12}}{4\pi R_{12}^2} = \frac{I\left[z\,r\,d\varphi\,\hat{a}_r + r^2\,d\varphi\,\hat{a}_z\right]}{4\pi\sqrt{r^2+z^2}\,\left(\sqrt{r^2+z^2}\right)^2}$$

$\hat{a}_r$ components $a_r^{\wedge}$ & $\hat{a}_z$.

...mmetry all $\hat{a}_r$ components

... so $\bar{H}$ will be along $\hat{a}_z$ dirn.



$$\frac{... r\, \hat{a}_r + r^2 \hat{a}_z] d\phi}{4\pi (r^2+z^2)^{3/2}}$$

$$\left\{ \int \frac{z r\, d\phi}{(r^2+z^2)^{3/2}} \hat{a}_r + \int_{\phi=0}^{2\pi} \frac{r^2 \hat{a}_z\, d\phi}{(r^2+z^2)^{3/2}} \right\}$$

$$0$$

$$\frac{...}{...^{3/2}} \hat{a}_z = \frac{I r^2 \hat{a}_z}{4\pi (r^2+z^2)^{3/2}} \int_{\phi=0}^{2\pi} d\phi$$

$$... \phi]_0^{2\pi} = \frac{I r^2 \phi \pi \hat{a}_z}{2\, 4\pi (r^2+z^2)^{3/2}}$$

$$\boxed{... \hat{a}_z \ A/m}$$

... circular loop

... of pt P along the axis.

... at centre of circular loop.

$$\boxed{\frac{...}{...^{3/2}} \hat{a}_z = \frac{I}{2r} \hat{a}_z \ A/m}$$

Ampere's Circuital Law

(similar to gauss law)

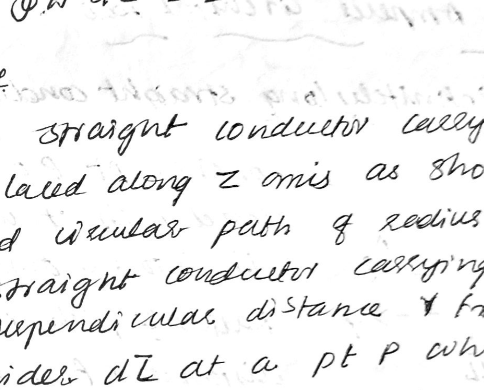
The line integral of magnetic field intensity $\bar{H}$ around a closed path is exactly equal to the direct current enclosed by that path.

$$\oint H \cdot dL = I$$

Proof

long straight conductor carrying direct current I placed along z axis as shown. Consider a closed circular path of radius r which encloses the straight conductor carrying DC I. Pt P is at perpendicular distance r from conductor. Consider $d\bar{L}$ at a pt P which is in $\hat{a}_\phi$ dirⁿ tangential to circular path at pt P.

$$d\bar{L} = r\, d\phi\, \hat{a}_\phi$$



$$d\bar{L} = r\, d\phi\, \hat{a}_\phi$$

$$\bar{H} = \frac{I}{2\pi r} \hat{a}_\phi \quad \text{(using Biot savart law)}$$

$$\bar{H} \cdot d\bar{L} = \frac{I}{2\pi r} \hat{a}_\phi \cdot r\, d\phi\, \hat{a}_\phi = \frac{I}{2\pi r} r\, d\phi$$

$$= \frac{I}{2\pi} d\phi$$

$d\phi = \frac{I}{2\pi}[\phi]_0^{2\pi}$

$= \frac{I}{}$

losing DC I, must be a

hape. The path considered is

s circuital law

y long straight conductor

consider pt P on the closed path at which $\bar{H}$ is to be obtained.

Radius of path r, P ⊥r distance r from conductor.

ent in $\hat{a}_\phi$ direcn. consider

$r d\phi \, \hat{a}_\phi$

$d\bar{l} = r d\phi \, \hat{a}_\phi$

$\phi \, \hat{a}_\phi = H_\phi r d\phi$

law.

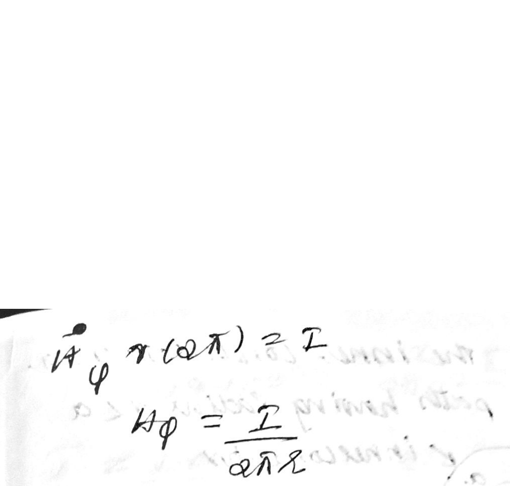
$d\phi = I$

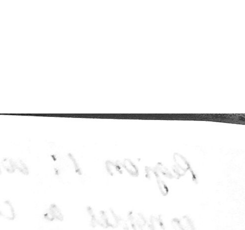
$H_\phi r(2\pi) = I$

$H_\phi = \frac{I}{2\pi r}$

$$\boxed{\bar{H} = H_\phi \hat{a}_\phi = \frac{I}{2\pi r}\hat{a}_\phi \text{ A/m}}$$

2) $\bar{H}$ due to co-axial cable





Consider a co-axial cable, its inner conductor is solid with radius a, carrying DC I. Outer conductor is in the form of concentric cylinder whose inner radius is b and outer radius is c. This cable is placed along z axis. The I is uniformly distributed in the inner conductor while I is uniformly distributed in the outer conductor.

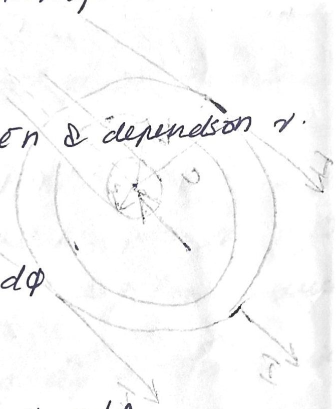
Inner & outer conductor is filled with dielectric. Calculation of $\bar{H}$ divided corresponding to various regions of cable.

inner conductor $r<a$.
having radius $r<a$
innerconductor

path

enclosed $\pi r^2 \; m^2$
& I through the area πa^2

I



diren & dependson r.

$\rightarrow r\, d\phi$

$\hat{a_\phi}$

$\hat{a_\phi} = H_\phi r d\phi$

circuital law

I

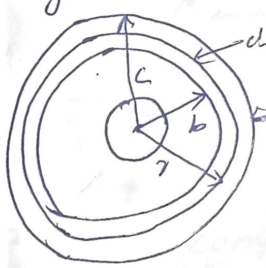
$$H_\phi = \frac{r^2}{2\pi r a^2} I = \frac{r}{2\pi a^2} I$$

$$\bar{H} = \frac{Ir}{2\pi a^2} \hat{a_\phi} \; A/m. \qquad r<a \text{ within conductor.}$$

Region 2: within $a<r<b$ consider a circular path which encloses the inner conductor carrying D.C I. (Infinitely long straight conductor)

$$\bar{H} = \frac{I}{2\pi r} \hat{a_\phi} \; A/m.$$

Region 3: $b<r<c$



$$I' = \frac{\pi(r^2-b^2)}{\pi(c^2-b^2)}(-I)$$

$$= -\frac{(r^2-b^2)}{(c^2-b^2)} I$$

$I'' = I \rightarrow$ current in innerconductor.

$$I_{enc} = I' + I'' = -\frac{(r^2-b^2)}{(c^2-b^2)} I + I$$

$$= I\left[1 - \frac{(r^2-b^2)}{(c^2-b^2)}\right] = I\left[\frac{c^2-r^2}{c^2-b^2}\right]$$

$$\oint H \cdot dl = I_{enc}$$

$$\bar{H} = H_\phi \hat{a_\phi} \quad \& \quad \bar{dl} = r\,d\phi\, \hat{a_\phi}.$$

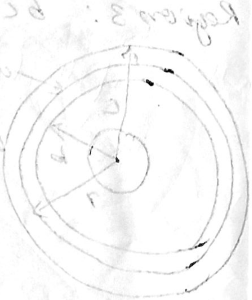
$$H \cdot dl = H_\phi r\, d\phi$$

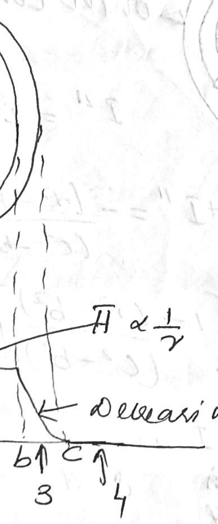
$$\int_{\phi=0}^{2\pi} H_\phi r\, d\phi = I_{enc}$$

$\frac{c^2-r^2}{c^2-b^2}\Big] a_\phi$

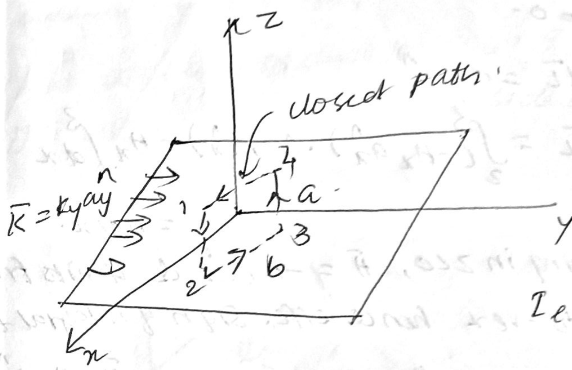
both conductors both current

$= 0$ A/m.





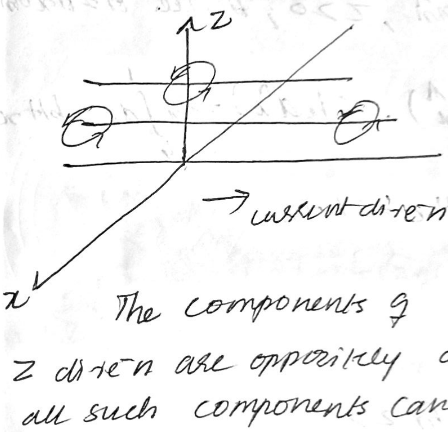
$\bar{H}$ due to infinite sheet of current



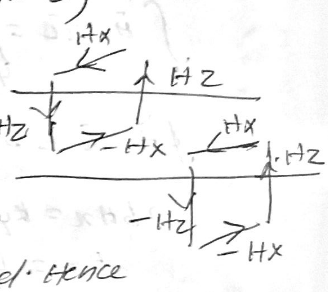
current flowing in distance b is $k_y b$.

$\bar{K}$ is the surface current density.

$I_{enc} = K_y b.$



(Right hand thumb rule)



The components of $\bar{H}$ in Z direction are oppositely directed. Hence all such components cancel each other.

As current is flowing in y direction, $\bar{H}$ can't have component in y direction.

$\bar{H} = H_x \hat{a_x}$ for $z > 0.$

$= -H_x \hat{a_x}$ for $z < 0.$

$\oint H \cdot dL = I_{enc}$

path 1-2-3-4-1

for path 1-2 $\bar{dL} = dz \hat{a_z}$

3-4 $\bar{dL} = dz \hat{a_z}$

$\hat{a_x} \cdot \hat{a_n} = 0.$

$\hat{a_n}$

$H_x\, \hat{a_x}) \cdot (dx\, \hat{a_n}) = H_x \int_2^3 dx$

$= 6H_x.$

$0, \ \bar{H} = -H_x\, \hat{a_n}$ & limits from

hence effe. sign of integral eve.

$\hat{a_n}, \ z>0;$ Hence $\bar{H} = H_x\, \hat{a_n}$

$). \ (dx\, \hat{a_x}) = H_x \int_4^1 dx = 6H_x$

$x = 26H_x$

for $z>0.$

$z<0.$

$\times \hat{a_N}.$

↳ unit vector normal from the current sheet to pt at which H is obtained.

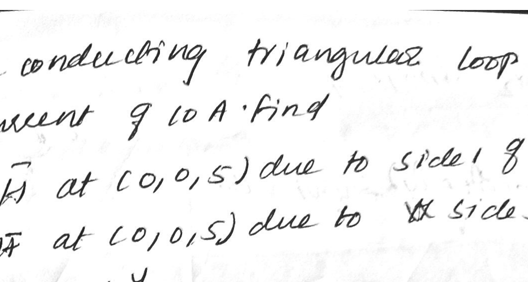
? The conducting triangular loop in the fig carries current of 10 A. find

i) $\bar{H}$ at $(0,0,5)$ due to side 1 of the loop

ii) $\bar{H}$ at $(0,0,5)$ due to side 3 of the loop.

i) $\bar{H} = \dfrac{I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1)\, \hat{a_\phi}.$

$\dfrac{-2\hat{a_x} + 5\hat{a_z}}{\sqrt{29}}$



$\sin\alpha_1 = 0$

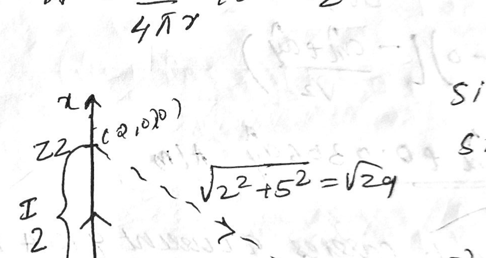
$\sin\alpha_2 = \dfrac{2}{\sqrt{29}}$

$\hat{a_\phi} = \hat{a_z} \times \hat{a_x}$

Here $\hat{a_n} \times \hat{a_z} = -\hat{a_y}$

$\bar{H} = \dfrac{10}{4\pi \times 5}\left(\dfrac{2}{\sqrt{29}} - 0\right)(-\hat{a_y}) = -0.059\, \hat{a_y}$ A/m.

ii)



$\begin{vmatrix} a_x & a_y & a_z \\ 2 & 0 & 0 \\ -2 & 0 & 5 \end{vmatrix}$

$\dfrac{-a_y(10) + a_z}{\sqrt{29}}$

$\alpha_1) \hat{a}_\varphi$

$a_x \; a_y \; a_z$

$\hat{a_y}) \rightarrow$ unit vector

$= -\frac{(-\hat{a_y} + \hat{a_x})}{\sqrt{2}} = \frac{-a_x + \hat{a_y}}{\sqrt{2}}$

$= 0$

$= \frac{\sqrt{2}}{3\sqrt{3}}$

$)\left(-\frac{\hat{a_x} + \hat{a_y}}{\sqrt{2}}\right)$

$+ 0.0306 \hat{a_y}$ A/m

carries a current of 10A in the z axis. Find the magnetic point P(1,2,3) due to this filament

b) z = 0 to 5 m c) z = ∞

a) $\bar{H}$ due to infinitely long straight conductor.

$\bar{H} = \frac{I}{2\pi r} \hat{a_\varphi}$ P(1,2,3) I = 10A

placed in z axis

$r = \sqrt{x^2 + y^2} = \sqrt{1+4} = \sqrt{5}$ m

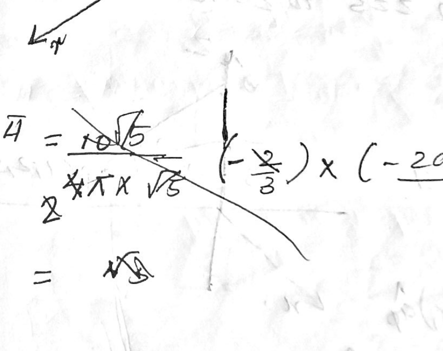
$a_x \; a_y \; a_z$

$\bar{H} = \frac{10}{2\pi \times \sqrt{5}} \times \hat{a_\varphi} = 0.7117 \hat{a_\varphi}$ A/m

$\hat{a_z} \times \frac{(a_x + 2a_y)}{\sqrt{5}} = \frac{a_y + 2(-\hat{a_x})}{\sqrt{5}}$

$\bar{H} = -0.63\hat{a_x} + 0.31\hat{a_y}$ A/m

b)



Here r remains the same & $\hat{a_\varphi}$ also same.

$\bar{H} = \frac{I}{4\pi r}(\sin\alpha_2 - \sin\alpha_1)\hat{a_\varphi}$

~~$\alpha_2 = 90, \sin\alpha_2 = 1$~~

$\sin\alpha_1 = 0$

$\sin\alpha_1 = 0.$

$\sin\alpha_2 = \frac{2}{3}$

$\bar{H} = \cancel{\frac{10\sqrt{5}}{2 \times 4\pi \times \sqrt{5}} \left(-\frac{2}{3}\right) \times \left(-\frac{2a_x + a_y}{\sqrt{5}}\right)}$

$=$

$\cancel{\sin\alpha_1 = \frac{3}{\sqrt{4+9}}}$ $\cancel{1+4}$

P(1,2,3) $\cancel{= 0.832}$

$\longrightarrow y$ $\cancel{\sin\alpha_2 = \frac{2}{\sqrt{8}}}$

$\sin\alpha_1 = \frac{3}{\sqrt{(\sqrt{5})^2+3^2}} = 0.802$

$\sin\alpha_2 = \frac{2}{\sqrt{(\sqrt{5})^2+2^2}} = \frac{2}{3}$

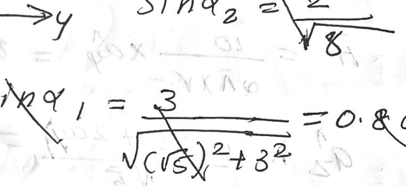
owpt P)

$= 0.66$

$0.801)\left(\frac{-2\hat{a_x} + \hat{a_y}}{\sqrt{5}}\right)$

$.233\,\hat{a_y}$

z5 to z = 0.



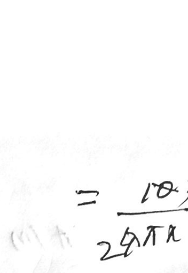
$\hat{a_\varphi}$

$= \frac{10 \times}{2\cancel{4}\pi \times \sqrt{5}} \times \frac{\sqrt{?}}{50} \times \left(\frac{-2\hat{a_x} + \hat{a_y}}{\sqrt{5}}\right)$

0.34

$= \frac{0.34}{2\pi}(-2\hat{a_x} + \hat{a_y}) = -108\,\hat{a_x} + 0.054\,\hat{a_y}$ A/m

? ~~A circular loop located at $x^2+y^2 = 9, z=0$ carries a direct current of 10A along $\hat{a_\varphi}$. Determine $\bar{H}$ at (0,0,4) & (0,0,−4)~~



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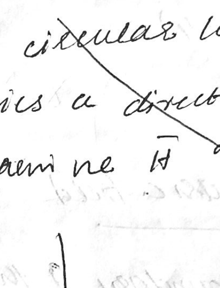


$\bar{H} = \frac{I r^2}{2(z^2+r^2)^{3/2}}\,\hat{a_z}$

$I = 10A$

$r = 3$

$z = 4$

$^n = \underline{0.36\, a_z^n}$ A/m

$z = -4$ no charge for

$a_y^n \times a_\phi^n$). There is nocharg

t $(0, 0, -4)$

A/m

s for Electric field

at the boundary of the 2

passes from one media

boundary cond^ns.

The following relations are

$\oint_l \bar{E} \cdot d\bar{l} = 0$

→ Normal to boundary

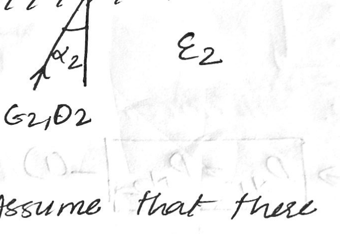
‖ to boundary

1) Dielectric Boundary cond^ns

consider boundary b/w 2 dielectric with different properties

$\epsilon_1 = \epsilon_0 \epsilon_{r1}$ & $\epsilon_2 = \epsilon_0 \epsilon_{r2}$

Lines of $\bar{E}$ & $\bar{D}$ will be refracted when passing from media to media,



D_1, D_2 → flux densities

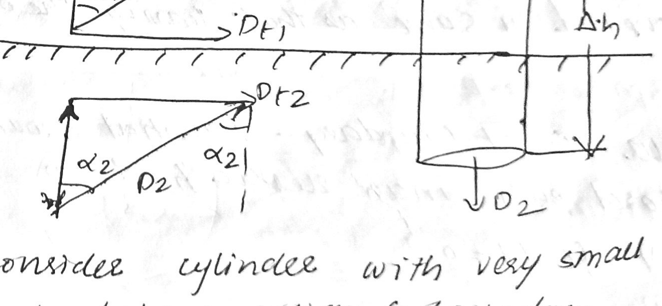
E_1, E_2 → field intensities in respective media.

$D_1 = \epsilon E_1 = \epsilon_0 \epsilon_{r1} E_1$

$D_2 = \epsilon E_2 = \epsilon_0 \epsilon_{r2} E_2$.

Assume that there is no free charge on the interface b/w the dielectrics.

ie $\rho_s = 0$.



Consider cylinder with very small thickness enclosed in a portion of boundary

$\oint D\cdot ds = Q$

-0

$\bar{D}\cdot d\bar{s} = 0$.

al

$\bar{D}\cdot d\bar{s} = -D_{N_2} ds$,

$\Rightarrow \boxed{D_{N_1} = D_{N_2}}$ —(1)

t of flux density $\bar{D}$ is

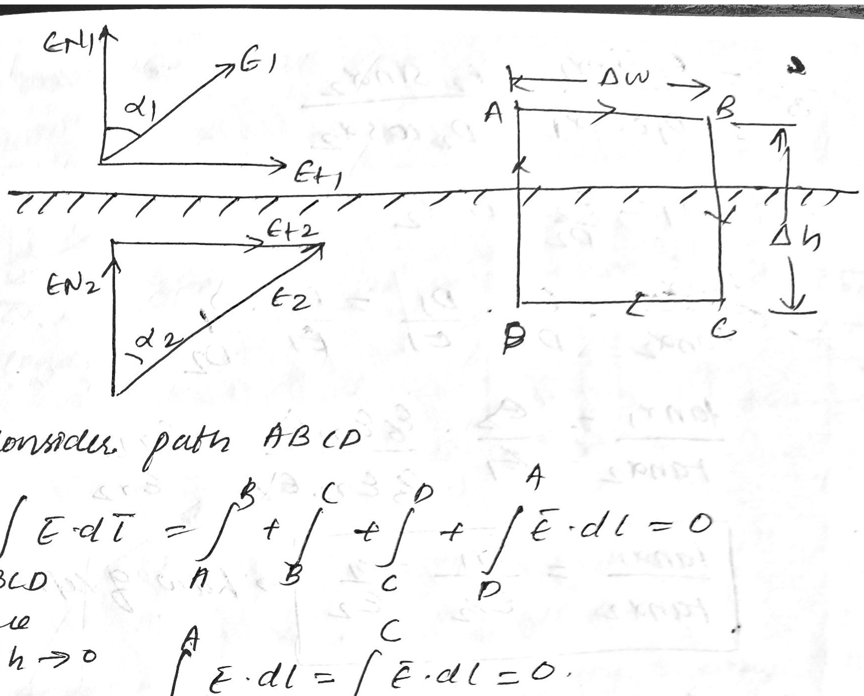
boundary b/w 2 perfect

displacement flux ent.

as that leaving the other

dary. Potential around

n electric field $= 0$



Consider path ABCD

$$\int_{ABCD} \bar{E}\cdot d\bar{l} = \int_A^B + \int_B^C + \int_C^D + \int_D^A \bar{E}\cdot dl = 0$$

since $\Delta h \to 0$

$$\int_D^A E\cdot dl = \int_B^C \bar{E}\cdot dl = 0.$$

$$\int_{ABCD} E\cdot dl = \int_A^B + \int_C^D E\cdot dl = 0.$$

$E_{t_1}\Delta w - E_{t_2}\Delta w = 0 \Rightarrow \boxed{E_{t_1} = E_{t_2}}$ —(2)

Tangential components of fields E_1 & E_2 at the boundary are equal.

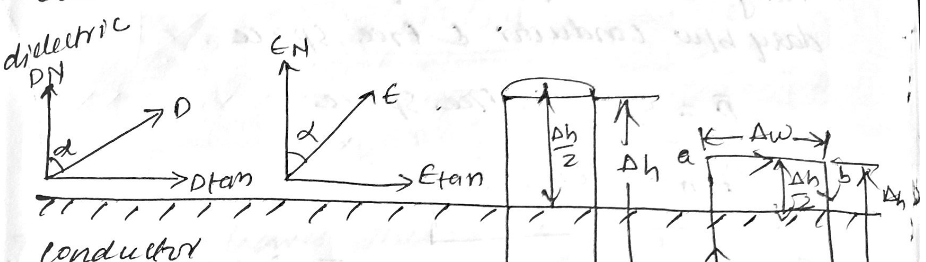
From (1) $D_{N_1} = D_{N_2}$

$D_1 \cos\alpha_1 = D_2 \cos\alpha_2$ —(3)

From (2) $E_{t_1} = E_{t_2}$

$E_1 \sin\alpha_1 = E_2 \sin\alpha_2$ —(4)

Conductor free space boundary



Consider path ABCDA

$$\oint_{ABCDA} E\cdot dl = \int_A^B + \int_B^C + \int_C^D + \int_D^A \bar{E}\cdot d\bar{l} = 0$$

Rectangle ABCD is placed in such a way that half of it is in the conductor, & remaining half in free space.

portion CD in the conductor where $\bar{E} = 0$

$$\int_C^D \bar{E}\cdot dl = 0$$

$$\int_A^B E\cdot dl = E_{tan}\,\Delta w \qquad \int_B^C E\cdot dl = -E_N \frac{\Delta h}{2}$$

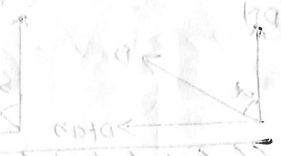
$$\int_D^A \bar{E}\cdot d\bar{l} = E_N \frac{\Delta h}{2}$$

$$\oint_{ABCDA} E\cdot dl = E_{tan}\,\Delta w - E_N\frac{\Delta h}{2} + 0 + E_N\frac{\Delta h}{2} = 0.$$

$$E_{tan}\,\Delta w = 0$$

$$\boxed{E_{tan} = 0}$$

... of E is zero at the boun-
... free space.
... space



... of D is zero at boundary
... space.
... surface in the form of a right
... ht Δh & placed in such
... the conductor & remaining
... e.

$= Q$.

... of cylinder gaussian surface
... $D = 0$

... o.

$\Delta S = Q$.

... arge exists in the form of ρ_s

$Q_s = \rho_s \Delta S$

$D_N \Delta S = \rho_s \Delta S$

$\boxed{D_N = \rho_s}$

$\epsilon E_N = \rho_s$

$\boxed{E_N = \dfrac{\rho_s}{\epsilon}}$

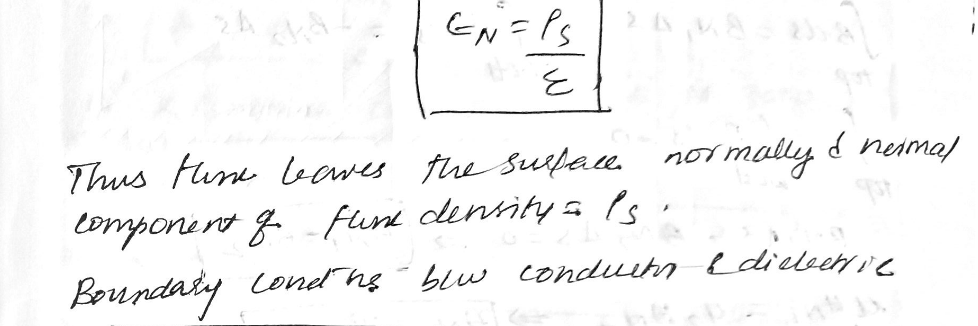
Thus flux leaves the surface normally & normal component of flux density = ρ_s.

Boundary cond'ns b/w conductor & dielectric

1. $D_{tan} = E_{tan} = 0$
2. $D_N = \rho_s$
3. $E_N = \dfrac{\rho_s}{\epsilon}$

Boundary conditions for Magnetic field

consider an interface b/w two magnetic media with relative permeabilities μ_1 & μ_2.



for normal component

Gauss law $\oint B \cdot ds = 0$

$$\int_{top} + \int_{bott} + \int_{later} = 0$$

$\oint \vec{B}\cdot d\vec{s} = -B_{N_2}\,\Delta S$

$\Rightarrow \boxed{B_{N_1} = B_{N_2}}$

$\boxed{H_{N_2} = \frac{\mu_1}{\mu_2} H_{N_1}}$

is continuous but normal comp

ntinuous by the ratio μ_1/μ_2

aw

$\int_C^D + \int_D^A \vec{H}\cdot d\vec{l}$

$\cdot d l = 0$

$\int_C^D \vec{H}\cdot dl = -H_{t_2}\,\Delta w$

$\int_C^D H\cdot dl$

$\Delta w = (H_{t_1} - H_{t_2})\,\Delta w$

$= k\,\Delta w$

k is the component of surface current normal to the plane of closed path

$$\boxed{k = H_{t_1} - H_{t_2}}$$

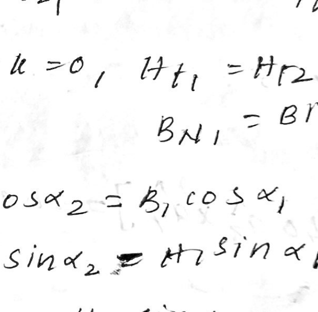
If K (surface current density) is zero

$$\boxed{H_{t_1} = H_{t_2}}$$

$$\frac{B_{t_1}}{\mu_1} = \frac{B_{t_2}}{\mu_2} \Rightarrow \boxed{\frac{B_{t_1}}{B_{t_2}} = \frac{\mu_1}{\mu_2} = \frac{\mu_{r_1}}{\mu_{r_2}}}$$

med 1

med. 2



Let the fields make angles α_1, α_2 with the normal to interface

if $k = 0$, $H_{t_1} = H_{t_2}$

$B_{N_1} = B_{N_2}$

$B_2 \cos\alpha_2 = B_1 \cos\alpha_1$ — (1)

$H_2 \sin\alpha_2 = H_1 \sin\alpha_1$ — (2)

$$\frac{(2)}{(1)} = \frac{H_2 \sin\alpha_2}{B_2 \cos\alpha_2} = \frac{H_1 \sin\alpha_1}{B_1 \cos\alpha_1}$$

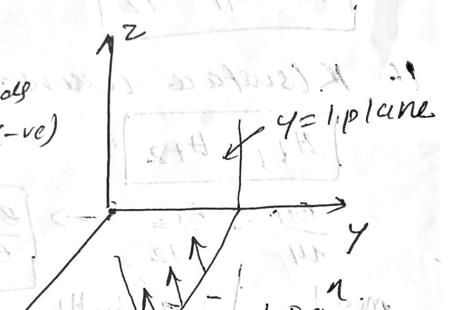
$$\frac{H_2}{B_2}\tan\alpha_2 = \frac{H_1}{B_1}\tan\alpha_1$$

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{H_2}{B_2} \times \frac{B_1}{H_1}$$

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{H_2}{H_1} \times \frac{\mu_1 H_1}{\mu_2 H_2} = \frac{\mu_1}{\mu_2}$$

$$\boxed{\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\mu_{r_1}}{\mu_{r_2}}}$$

s current density $\bar{K} = 40\,\hat{a}_z$ A/m.
& B (1, 5, -2)



olg
-ve)
$y = 1$, plane
$\bar{K} = 40\,\hat{a}_z$

A/m.

as
$y = 5$

$\frac{1}{2}[40\,\hat{a}_z \times \hat{a}_y] = -20\,\hat{a}_x$ A/m.

I_c
ductor due to flow of elec^ns
etric potential.
nsity
pt form of ohms law

→ current in a conductor when electric field remains const. w.r.t time

<u>Displacement current I_D</u>

→ current due to changing electric field is called displacement current. It is non zero under all time varying electric field. In case where electric field doesn't change with time (steady electric field) in conducting wire I_D may be $= 0$

→ I_D is the current which exist b/w 2 plates of capacitor when there is time varying electric field.

Outside capacitor plates

$I_c = i \quad \& \quad I_D = 0$

Inside capacitor plates

$I_D = i \quad I_c = 0$

$I_D = C\frac{dV}{dt} \qquad C = \frac{\varepsilon A}{d}, \quad V = E d$

$I_D = \frac{\varepsilon A}{d}\frac{d}{dt}(E d) = \frac{\varepsilon A}{\cancel{d}}\cancel{d}\frac{dE}{dt} = \varepsilon A\frac{dE}{dt}$

$E = \frac{D}{\varepsilon}$

$I_D = \varepsilon A\frac{d}{dt}(D/\varepsilon) = \cancel{\varepsilon} A \cdot \frac{1}{\cancel{\varepsilon}}\frac{dD}{dt} = A\frac{dD}{dt}$

Displacement current density

$\boxed{J_D = \frac{I_D}{A} = \frac{\partial D}{\partial t}}$

Vector form

$\boxed{\bar{J}_D = \frac{\partial \bar{D}}{\partial t}}$

$\frac{\partial}{\partial t} = j\omega$

$\boxed{= \frac{\sigma}{\omega\varepsilon}}$

...du'n current is very large
...ment current (σ large)
...m; displacement current
...o condu'n current
...edium is conductor)
...medium is dielectric)
...nds on frequency. At low
...1 of medium is condu-
...< 1, med. is dielectric.

Angle b/w J_c & J_D

If $E = E_m \sin\omega t$, then $D = \varepsilon E = \varepsilon E_m \sin\omega t$

$$J_D = \frac{\partial D}{\partial t} = \frac{\partial}{\partial t}(\varepsilon E_m \sin\omega t)$$
$$= \omega \varepsilon E_m \cos\omega t$$

$$J_c = \sigma E = \sigma E_m \sin\omega t$$

Angle b/w J_c & J_D is $90°$

$\tan\theta = \frac{J_c}{J_D} = \frac{\sigma}{\omega\varepsilon}$; $\tan\theta$ is known as tangent loss or loss tangent of dielectric. θ is called loss angle.

Conduction current	Displacement current
1) caused by electric charges in motion	1) caused due to bound charges
2) $I_c = \oint_L \bar{H} \cdot d\bar{l} = \oiint_S \bar{J} \cdot d\bar{s}$	2) $I_d = \oint \bar{H} \cdot d\bar{l} = \oiint_S \bar{J}_D \cdot d\bar{s}$ $= \oiint \frac{\partial \bar{D}}{\partial t} \cdot d\bar{s}$
3) $\bar{J}_c = \sigma \bar{E}$	3) $\bar{J}_D = \frac{\partial \bar{D}}{\partial t} = j\omega\varepsilon\bar{E}$
4) freq indept.	4) freq dept.
5) Dominant in conductor	5) Dominant in dielectric

Continuity equation

It is based on law of conservation of charges. It states that "charges can neither be created nor be destroyed"

surface 'S' with current
the total current I crossing

g charge there must be
amt of +ve charges inside
ne the outward rate of
t balanced by the rate of
side the closed surface i.e

$\boxed{\frac{dQ}{dt}}$ —(1)

side the closed surface 'S',
decrease in charge.

ing

—(2)

wn as integral form of conti-

$\bar{J}$) dV (using divergence theorem)

$\frac{dQ}{dt}$

$$\iiint_V (\nabla \cdot \bar{J})\, dv = -\frac{d}{dt}\left[\iiint_V \rho_v\, dv\right]$$

$$= \iiint_V -\frac{d}{dt}\rho_v\, dv.$$

$$\iiint_V (\nabla \cdot \bar{J})\, dv = \iiint_V -\frac{\partial}{\partial t}\rho_v\, dv$$ → for constant surface the derivative becomes partial derivative

$$\boxed{\nabla \cdot \bar{J} = -\frac{\partial}{\partial t}\rho_v}$$

→ point form / differen' form of continuity eqn

Continuity eqn states that current or charge per second diverging ~~per~~ from a small volume per unit volume is equal to the time rate of decrease of charge per unit volume at every pt.

For steady current $\frac{\partial \rho_v}{\partial t} = 0$

$$\Rightarrow \nabla \cdot \bar{J} = 0.$$

Maxwell's Equations:-

1) from modified form of Ampere's circuital law:-

From Ampere circuital law

$$\oint_L \bar{H} \cdot \bar{dl} = I_{enc} = \oiint_S \bar{J} \cdot \bar{ds} \text{ (Integral form)}$$

$$\nabla \times \bar{H} = \bar{J} \quad \text{—(a)} \quad \text{point form.}$$

both sides of eqn

y curl is zero

zero.

varying field bcoz the

time varying current is

ero for time varying current

ent for time varying fields

n

(2)

$\nabla \cdot \bar{D} = \rho_v$, subst in (2)

$\nabla \cdot [\bar{J} + \frac{\partial \bar{D}}{\partial t}] = 0$

ivergence of $\bar{J} + \frac{\partial \bar{D}}{\partial t} = 0$

Modified Ampere Circuital law

$$\boxed{\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}} \rightarrow \text{Point form}$$

$$\boxed{\oint_L \bar{H} \cdot d\bar{l} = \iint_S \left(\bar{J} + \frac{\partial \bar{D}}{\partial t}\right) \cdot d\bar{s}} \rightarrow \text{Integral form.}$$

From Faraday's Law

Faraday's I Law :- whenever flux linked with a coil changes an emf is induced in it

Faraday's II Law :- The magnitude of induced emf is equal to the rate of change of flux linkage

$$e = -N\frac{d\phi}{dt} \quad (\text{-ve sign Lenz's Law})$$

consider a closed circuit made of a single turn loop. If magnetic field is $\perp$ to the plane of the loop & is varying an emf is induced in the conductor & a current flows in the closed ckt.

By faraday's Law

$$e = -\frac{d\phi}{dt} \quad (N = 1) \quad \text{— (1)}$$

The emf induced is also given by

$$e = \oint_L \bar{E} \cdot d\bar{l} \quad \text{— (2)}$$

equating (1) & (2)

$$\oint_L \bar{E} \cdot d\bar{l} = -\frac{d\phi}{dt}$$

Total flux through a circuit

$$\phi = \oiint_S \bar{B} \cdot d\bar{s}$$

$$\oint_L \bar{E} \cdot d\bar{l} = \frac{-d}{dt}\left[\oiint_S \bar{B} \cdot d\bar{s}\right]$$

$$= -\oiint_S \frac{\partial B}{\partial t} \cdot d\bar{s} \quad — (3)$$

$$\oint \bar{E} \cdot d\bar{l} = \oiint_S (\nabla \times \bar{E}) \cdot d\bar{s} \quad — (4) \text{ (Stoke's theorem)}$$

From (3) & (4) we get

$$\oiint_S (\nabla \times \bar{E}) \cdot d\bar{s} = -\oiint_S \frac{\partial \bar{B}}{\partial t} d\bar{s}$$

$$\boxed{\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}}$$ → point form / differential form

$$\boxed{\oint_L \bar{E} \cdot d\bar{l} = -\oiint_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{s}}$$ → Integral form.

From Gauss Law for electric field

Integral form of Gauss law is

$$\oiint_S \bar{D} \cdot d\bar{s} = Q$$

but $Q = \iiint_V \rho_v \, dv$ — (1)

Applying divergence theorem,

$$\oiint \bar{D} \cdot d\bar{s} = \iiint_V (\nabla \cdot \bar{D}) \, dv \quad — (2)$$

Comparing (1) & (2):

$$\boxed{\nabla \cdot \bar{D} = \rho_v}$$ → Point form

$$\boxed{\oiint \bar{D} \cdot d\bar{s} = \iiint_V \rho_v \, dv}$$ → Integral form

From Gauss Law for magnetic field

Integral form of Gauss Law

$$\boxed{\oiint \bar{B} \cdot d\bar{s} = 0}$$

By divergence theorem

$$\oiint \bar{B} \cdot d\bar{s} = \iiint_V (\nabla \cdot \bar{B}) \, dv = 0$$

$$\boxed{\nabla \cdot \bar{B} = 0}$$ → Point form of Gauss Law.

Magnetic scalar & vector potentials

Scalar Magnetic potential

V_m → scalar magnetic potential, then it must satisfy

$$\nabla \times \nabla V_m = 0$$

$$\bar{H} = -\nabla V_m$$

ie $\nabla \times \bar{H} = 0$

ie, $\bar{J} = 0$

...ic potential V_m can be

...free region where $\bar{J} = 0$

only for $\bar{J} = 0$

...$\bar{I}$

... for scalar Magnetic Potential

...orem

... $dv = 0$

...$(\mu_0 \bar{H}) = 0$ but, $\mu_0 \neq 0$

...$\cdot(-\nabla V_m) = 0$ $\bar{H} = -\nabla V_m$

$\boxed{\text{for } \bar{J} = 0}$

...n for scalar magnetic

...r to laplace's eqn for

...al $\nabla^2 V = 0$

Vector Magnetic Potential

Denoted as $\bar{A}$ & measured in wb/m.

$\nabla \cdot (\nabla \times \bar{A}) = 0$ $\bar{A} \rightarrow$ vector magnetic potential

$\nabla \cdot \bar{B} = 0$

$$\boxed{\bar{B} = \nabla \times \bar{A}}$$

Curl of vector magnetic potential is the flux density

$\nabla \times \bar{H} = \bar{J}$; $\nabla \times \frac{\bar{B}}{\mu_0} = \bar{J}$; $\bar{B} = \mu_0 \bar{H}$

$\nabla \times \bar{B} = \mu_0 \bar{J}$ $\bar{B} = \nabla \times \bar{A}$

$\nabla \times \nabla \times \bar{A} = \mu_0 \bar{J}$

using vector identity

$\nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A} = \mu_0 \bar{J}$

$$\bar{J} = \frac{1}{\mu_0}[\nabla \times \nabla \times \bar{A}] = \frac{1}{\mu_0}[\nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}]$$

Thus if vector magnetic potential is known then current density $\bar{J}$ can be obtained.

? At the boundary b/w glass & air the lines of electric field in air makes an angle 40° with normal to boundary. If flux density in air is 0.25 $\mu C/m^2$. Determine the orientation & magnitude of electric flux density in glass. Also find electric field intensity in glass & air.

Glass 

ϵ_1, D_1

$\alpha_2 = 40^\circ$

$\alpha_1 = ?$ $D_1 = ?$

$E_1 = ?$ $E_2 = ?$

$\epsilon_{r1} = 4$ $\epsilon_{r2} = 1$

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$

$$\tan\alpha_1 = \frac{\epsilon_{r1}}{\epsilon_{r2}} \times \tan\alpha_2 = \frac{4}{1}\tan 40^\circ$$

$$= 3.356$$

$$\alpha_1 = \tan^{-1}(3.356)$$

$$= \underline{73.40}$$

$$D_2 = \epsilon_0 \epsilon_r E_2, \quad E_2 = \frac{D_2}{\epsilon_0 \epsilon_{r2}} = \frac{0.25 \times 10^{-6}}{8.854 \times 10^{-12} \times 1}$$

$$= 28235.82\ V/m.$$

$$D_{N1} = D_{N2}$$

$$D_1 \cos\alpha_1 = D_2 \cos\alpha_2$$

$$D_1 = D_2 \frac{\cos\alpha_2}{\cos\alpha_1} = 0.25 \times 10^{-6} \times \frac{\cos 40^\circ}{\cos 73.40^\circ}$$

$$= 6.706 \times 10^{-7}\ C/m^2$$

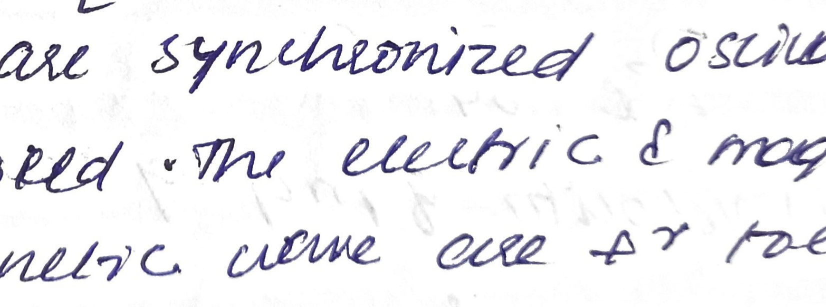
$\varepsilon_r, \varepsilon,$

ε_r

$\dfrac{\cdot 706 \times 10^{-7}}{\cdot 854 \times 10^{-12} \times 4}$

$18934 \cdot 94$ V/m

-4



es :- They are synchronized oscilations of
magnetic field. The electric & magnetic
electromagnetic wave are ⊥r to each
... the diren of propagation